

Status	Finished
Started	Tuesday, 3 March 2026, 4:47 PM
Completed	Tuesday, 3 March 2026, 6:17 PM
Duration	1 hour 30 mins
Grade	36.75 out of 48.00 (76.56%)

Information

Allowed Aids

During this exam, you are allowed to use the following aids:

- A pen
- Your own non-programmable calculator
- This calculator website: <https://yogeshiitm.github.io/scientific-calculator/>

Question 1

Correct

Mark 1.00 out of 1.00

What is meant by the term **spatial locality**?

Select one:

- ☐ The likelihood of accessing data that is already accessed.
- ☒ The likelihood of accessing data that is stored together. ✓

Your answer is correct.

The correct answer is: The likelihood of accessing data that is stored together.

Question 2

Correct

Mark 4.00 out of 4.00

Consider a hard disk with the following characteristics:

- 8 platters with 16 surfaces total
- 8192 ($= 2^{13}$) tracks per surface
- 1024 ($= 2^{10}$) sectors per track (this is a simplified model where all tracks have the same number of sectors)
- 1024 ($= 2^{10}$) bytes per sector
- 16384 ($= 2^{14}$) bytes per block
- 7200 rotations per minute
- 1 ms to start and stop the read/write head (1 ms for both operations together)
- 1 ms per 4000 cylinders that have to be passed by the read/write head

What is the overall capacity of the disk in Gigabytes (GB, 1 GB = 1024^3 Bytes)? (Please answer with a simple number. Do not use any units. Round the final result to the nearest integer.)

Answer: ✓

$16 * 2^{13} * 2^{10} * 2^{10} / 2^{30} \# 128$

The correct answer is: 128

Question 3

Incorrect

Mark 0.00 out of 6.00

Consider a hard disk with the following characteristics:

- The read/write head is initially at cylinder 36000.
- The average transfer time for a block is 0.33 ms.
- The average rotational latency is 4.17 ms.
- It takes 0.5 ms to start and 0.5 ms to stop the head (1 ms for both operations together).
- It takes 1 ms per 4000 cylinders that have to be passed by the head.

Requests arrive at the following cylinders and timestamps:

Cylinder of request	First time available (ms)
20000	0
60000	1
24000	2
48000	20
8000	30
16000	40

Requests are going to be processed according to the **elevator (SCAN)** algorithm. **Choose the processed requests in the correct order, together with their completion time (ms).**

Notes:

- Requests are named after their arrival (first available time). For example, if a request arrives at $10ms$, then it is called t_{10} .
- We assume that the disk head is initially stationary.
- Answer with a number only (do not use units). Use decimal place if needed (e.g., 5.2 or 7).
- Round the final values to one decimal place (e.g., $3.889 \rightarrow 3.9$).
- Completion time refers to the wall-clock time measured from the start of the schedule (0 ms). Therefore, it should never decrease in the table as they also incorporate the previous operations.

Request Time completed (ms)

t0	9.5	⊗
⊗		
t2	16	⊗
⊗		
t1	30.5	⊗
⊗		
t20	39	⊗
⊗		
t30	54.5	⊗
⊗		
t40	62	⊗
⊗		

Order: t2 -> t0 -> t20 -> t1 -> t40 -> t30

Calculation:

$$\begin{aligned}
 t_2 &= \text{abs}(36000 - 24000) / 4000 + 0.33 + 4.17 + 1 \\
 t_0 &= t_2 + \text{abs}(24000 - 20000) / 4000 + 0.33 + 4.17 + 1 \\
 t_{20} &= t_0 + \text{abs}(20000 - 48000) / 4000 + 0.33 + 4.17 + 1 \\
 t_1 &= t_{20} + \text{abs}(48000 - 60000) / 4000 + 0.33 + 4.17 + 1 \\
 t_{40} &= t_1 + \text{abs}(60000 - 16000) / 4000 + 0.33 + 4.17 + 1 \\
 t_{30} &= t_{40} + \text{abs}(16000 - 8000) / 4000 + 0.33 + 4.17 + 1
 \end{aligned}$$

Note: As we explained in the exercise and also in an announcement, you have to always check whether there is a request on the way of the disk head.

Question 4

Correct

Mark 1.00 out of 1.00

Row stores have high tuple reconstruction costs.

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question 5

Correct

Mark 2.00 out of 2.00

Consider the following relation R:

```
CREATE TABLE R ( employee_id INT, payment_id INT, amount INT, date DATE );
```

Assume that:

- Each field in a record consumes 4 bytes (integers as well as the date).
- Block size as well as page size is 32 bytes.
- No extra storage overhead is required for any storage model (i.e., page and block are same sized and contain purely data.)
- For each query in the questions below, the disk head is above the first block to be read.
- No further information about the schema exists, especially no info on indices and constraints.

What is the maximum number of records that R can contain such that the following query requires at most 5 block accesses? Assume the worst case, i.e., that all blocks of every accessed column must be read.

```
select employee_id from R
where date between start_date and end_date
```

N-ary storage model: ✓

Decomposition storage model: ✓

NSM: each block holds $32 / 16 = 2$ records. We can read five blocks --> 10 records. DSM: we need to scan all date entries. A block can contain $32 / 4 = 8$ date entries. If we have three date blocks, we also have to read three employee_id blocks (6 total), which exceeds the 5 block limit. With two date blocks (16 records) we need $2 + 2 = 4$ block accesses, which fits. Note: DSM (16) outperforms NSM (10) here because only 2 of 4 columns are accessed.

Question 6

Correct

Mark 1.00 out of 1.00

We have a query plan which has a skewed access pattern, i.e., it accesses a particular sections of data more frequently during the execution.

Given this workload, which caching strategy is more efficient?

Select one:

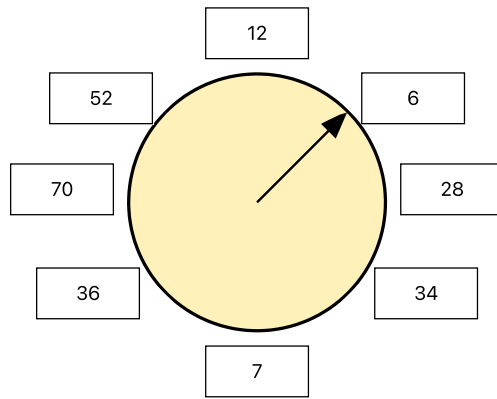
- ☒ LFU ✓
- ☐ LRU

Your answer is correct.

The correct answer is: LFU

Question 7
Partially correct
Mark 1.50 out of 4.00

Consider the CLOCK (a.k.a. Second Chance) algorithm. The system cache has space for 8 pages, and the current content of the cache is shown in the figure below.



5	Cached page: 5 Reference bit: 1
23	Cached page: 23 Reference bit: 0

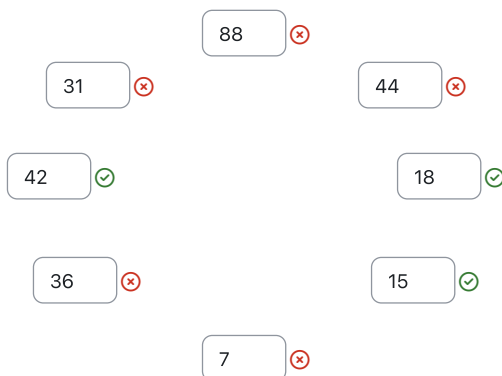
Assume that when a page is accessed and it is already in the cache, its reference bit is set to 1 (if it is 0, it is changed to 1; if it is 1, then it is left as 1). The clock hand does not move in such a case (i.e., it stays where it was before this access). The clock hand only moves when the accessed page is not in the cache and the cache is looking for a page to evict by moving clockwise. When a new page is brought into the cache its reference bit is set to 1.

Also assume that in the first figure; orange box indicates that the reference bit is set to 1, and yellow box indicates that reference bit is set to 0.

Consider the following sequence of page requests to the cache:

18, 15, 36, 6, 42, 3, 88, 44

Fill in the pages of the clock with regards to the page numbers after the last request has been completed (after all the pages in the series are accessed).



Question 8

Correct

Mark 2.00 out of 2.00

Following is the initial version of a cache with capacity of 5 pages. New pages are being added on the left-hand side of the cache, and the rightmost page is being evicted. If there is a cache hit, then no modification is done in the cache.

⇒ [23 , 88 , 34 , 3 , 15] ⇒

The Table below contains the pages to be accessed in the given order (from top to bottom). **For each page, decide whether it is a cache hit or cache miss. Assume that the cache eviction policy is first in first out (FIFO).**

Hit	Miss			
		22		
		23		
		15		
		3		
		6		

- .
- 22: Miss
- 23: Hit
- 15: Miss
- 3: Miss
- 6: Miss

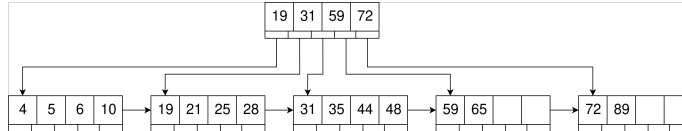
Question 9

Correct

Mark 5.00 out of 5.00

Consider the B⁺ tree below with the following properties:

- Each node (except the root) contains $2 \leq k \leq 4$ keys.
- Inner nodes: The keys in the subtree below the pointer p_i are less than the key k_i ; the keys in the subtree below the pointer p_{i+1} are greater or equal than the key k_i .
- Insert operations may only trigger node splits but not shifting of keys into bordering leaves.
- When a node is split, the middle key is moved to the parent node.
- Pointers $p_1, \dots, p_4$ in leaves point to row IDs. Pointer p_5 in leaves points to the next leaf.



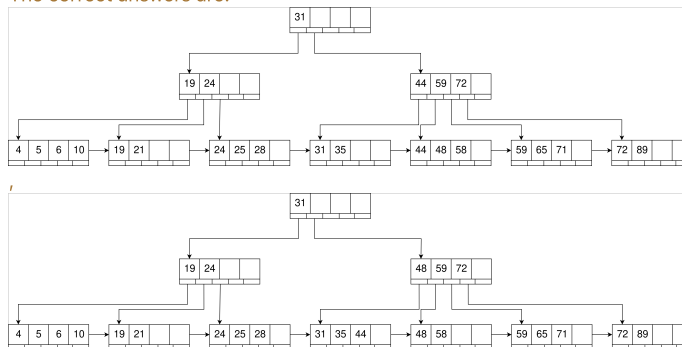
Which of the following choices represents the B⁺-Tree after insertion of the key 24, 71, 58 (in that order)?

Select one:

- ☐
- ☒
- ☐
- ☐

Your answer is correct.

The correct answers are:



Question 10

Correct

Mark 2.00 out of 2.00

Given the following properties of the data file, the disk, and index structures.

Data file:

- The data file contains 60 million tuples.
- Each tuple requires 256 bytes on disk.

Disk properties:

- The size of a disk block is 4096 bytes.

Index properties:

- The size of a search key is 4 bytes.
- The size of a pointer to a block is 8 bytes.
- Index entries do not span blocks on disk.

How many blocks are required for storing this second-level sparse index, if the first-level index is also sparse?

(Give your answer as a single integer.)

Answer: ✓

The correct answer is: 33

Question 11

Correct

Mark 2.00 out of 2.00

For the following relation, we are given the cardinality of the table and their attributes.

$R(a, b, c, d)$

- $T(R) = 420000$
- $V(R, a) = 35$
- $V(R, b) = 40$
- $V(R, c) = 20$
- $V(R, d) = 35$

What is the cardinality of the following selection?

$\sigma_{b \neq 64 \vee d < 36}(R) =$ ✓

The correct answer is: 413000

Question 12

Correct

Mark 3.00 out of 3.00

Given are four tables $R(a, e)$, $S(b, d)$, $T(c, e)$, and $U(a, d)$. Each relation has 1000 tuples. The relations have the following attribute domain cardinalities:

$R(a, e)$	$S(b, d)$	$T(c, e)$	$U(a, d)$
$V(R, a) = 20$			$V(U, a) = 400$
	$V(S, b) = 10$		
		$V(T, c) = 400$	
	$V(S, d) = 20$		$V(U, d) = 500$
$V(R, e) = 50$		$V(T, e) = 1000$	

Calculate the cardinality for the following two joins:

$T(S \bowtie U) =$ ✓

$T(T \bowtie R \bowtie U \bowtie S) =$ ✓

Question 13

Correct

Mark 1.00 out of 1.00

The Isolation property (of ACID) means that a transaction should complete all of its operations successfully

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question 14

Correct

Mark 3.00 out of 3.00

Consider the following schedule S :

$S: w_2(A); w_2(A); w_1(C); w_3(B); r_2(B); r_3(C); w_1(C); w_2(A);$

In the table below, indicate whether the listed edges exist in the precedence graph or not. (For example, if there exists an edge from T1 to T2 in the precedence graph, then the entry for T1→T2 should be Yes.)

Yes	No			
☐	☒	T1 → T2	☒	
☐	☒	T2 → T3	☒	
☒	☐	T3 → T2	☒	
☐	☒	T2 → T1	☒	
☒	☐	T3 → T1	☒	
☒	☐	T1 → T3	☒	

T1 → T2: No
 T2 → T3: No
 T3 → T2: Yes
 T2 → T1: No
 T3 → T1: Yes
 T1 → T3: Yes

Question 15

Incorrect

Mark 0.00 out of 2.00

Given is the following schedule.

$S_1: r_2(C); w_3(B); w_3(B); w_2(C); w_1(C); r_1(B); r_1(C); r_2(A);$

Is this schedule strict 2PL-compliant?

Note: Consider the schedule **as is**, i.e., you **do not** have to move individual operations around to find a compatible schedule.

- ☒ True ✗
- ☐ False

The schedule S_1 is 2PL-compatible as follows (locks are shown in bold):
S2(C) R2(C) **X3(B)** W3(B) W3(B) **X2(C)** W2(C) **S2(A)** **U2(C)** **X1(C)** W1(C)
U3(B) **S1(B)** R1(B) R1(C) R2(A) **U2(A)** **U1(C)** **U1(B)**

However, schedule S_1 is **not strict** 2PL-compatible because: Transaction 2 unlocked exclusive lock on C and executes R2(A) after that (in strict 2PL, exclusive locks are only released at the end of a transaction).

The correct answer is 'False'.

Question 16

Complete

Not graded

Consider the following transaction T1.

```
T1: BEGIN
  R(A)
  R(B)
  if A == 0 then B = B + 1
  W(B)
  COMMIT
```

Insert necessary locks/unlocks into correct positions in the transaction, so that T1 becomes 2PL compliant.

Note: If there is no need to insert a lock/unlock in that position, then select **no-action**.

T1

R(A)

R(B)

if A == 0 then B = B + 1

W(B)

COMMIT

Question 17

Complete

Not graded

To perform the ANALYSIS phase, we reconstruct the state of the database before the crash by loading the transaction table and dirty page table stored in the last successful checkpoint and scanning the contents of the UNDO/REDO log from the end of the last successful checkpoint.

- ☒ True
- ☐ False

The correct answer is 'False'.

Question 18

Partially
correctMark 5.25 out
of 6.00

Given below are the contents of an undo/redo log.

You are required to do two tasks in this question:

1. Fill out the transaction table and the dirty page table after the analysis pass below.
2. For each update action in the Undo / Redo Log at the beginning, mark if it is redone or undone.

Notes

- Each record contains a *log sequence number* (LSN), a *transaction id* (TID), the *previous LSN* (prevLSN), and the log record *action*.
- Update log records contain 4-tuples $\langle p, e, v, w \rangle$ where p is the updated page ID (PID), e the *updated element*, v the *old value*, and w the *new value*.
- Besides the Active Transaction Table and the Dirty Page Table you are also given a Table that specifies up to which LSN the data is persisted on disk.

Undo / Redo Log

LSN	TID	prevLSN	ACTION	Redo	Undo
1	1		BEGIN_TRANSACTION		
2	2		BEGIN_TRANSACTION		
3	2	2	UPDATE (1, r, "a", "A")	No	No
				✓	✓
4	2	3	UPDATE (2, s, "b", "B")	No	No
				✓	✓
5	1	1	UPDATE (2, t, "c", "C")	Yes	Yes
				✓	✓
6			CHECKPOINT		
7	2	4	COMMIT		
8	2	7	END TRANSACTION		
9	1	5	UPDATE (2, u, "d", "D")	Yes	Yes
				✓	✓

State at the checkpoint (i.e., the beginning of the recovery process)

Active Transaction Table Dirty Page Table

TID	lastLSN	PID	recLSN
1	5	2	5
2	4		

Information persisted on disk

PID	Page LSN
1	3
2	4

Transaction Table and Dirty Page Table after the analysis phase

Active Transaction Table Dirty Page Table

TID	lastLSN	PID	recLSN
1	9	1	9
	✓		✗

TID lastLSN

2

No Entry



PID recLSN

2

5



Analysis phase: Starting from the checkpoint state, scan the log forward from LSN 6.

- LSN 7 (T2 COMMIT): Remove T2 from the ATT.
- LSN 8 (T2 END): No further effect (T2 already removed).
- LSN 9 (T1 UPDATE P2): Update T1's lastLSN to 9. Page 2 is already in the DPT (recLSN = 5, unchanged).
- **Result:** ATT = {T1, lastLSN 9}. DPT = {P2 recLSN 5}.

Redo phase: Scan forward from $\min(\text{recLSN}) = 5$.

- LSN 3 (T2, P1): P1 is not in the DPT → skip. **Not redone.**
- LSN 4 (T2, P2): recLSN for P2 is $5 > 4$ → skip. **Not redone.**
- LSN 5 (T1, P2): recLSN = $5 \leq 5$, and pageLSN on disk = $4 < 5$ → **Redone.**
- LSN 9 (T1, P2): recLSN = $5 \leq 9$, and pageLSN on disk = $4 < 9$ → **Redone.**

Undo phase: Loser transactions: {T1}. Follow the prevLSN chain backward.

- LSN 9 (T1, P2): **Undone.** prevLSN = 5.
- LSN 5 (T1, P2): **Undone.** prevLSN = 1 (BEGIN) → stop.
- LSN 3, 4 belong to T2 (committed) → **Not undone.**

Question 19

Correct

Mark 1.00 out of 1.00

A major advantage of the shared-nothing architecture is that a single node failure does not impact system performance.

- ☐ True
- ☒ False ✓

In shared-nothing architectures, each node has its own CPU, memory, and storage. We partition data and distribute it across nodes, so that each node works on a specific part of data. If a node fails, then we have to take care of the recovery of that node (including the program data in its disk). Even if we have a replication for each node (this is not denoted in the question, and cannot be assumed), we still have to take care of the data copies for these replicas. Therefore, we cannot say that a failure does not impact the performance, and it cannot be a major advantage of the shared-nothing architectures.

The correct answer is 'False'.

Question **20**

Correct

Mark 2.00 out
of 2.00

A program takes 27 hours to complete when executed on a single processor. 80% of this program can be parallelized.

We want to execute this program on a system with 6 processors. **Calculate the new execution time of this program by using the theoretical speedup according to Amdahl's law.**

Note: Theoretical speedup according to Amdahl's law:

$$S = \frac{1}{(1 - P) + \frac{P}{N}}$$

Note 2: Answer with a simple number. Do not use any units. Round the **final result** to the nearest **integer**.

Answer:



`27 // (round( 1 / ( (1 - 0.8) + (0.8 / 6) ), 0)) # 9`

The correct answer is: 9